

CASE STUDY

Task for Course: DLBDSME01 – Model Engineering

CONTENT

1. Task.....	2
1.1 Task 1: Development of an Interpretable Model for Breast Cancer Prediction	3
1.2 Task 2: Development of a Demand Forecasting Model for Public Transport	4
1.3 Task 3: Development of a spam filter	5
2. Additional information for the evaluation of the case study	6
3. Tutorial Support	6

1. TASK

In this section, you can select one of the listed case studies to work on (see sections 1.1, 1.2, 1.3).

When working on your case study, please consider the task described in the respective case study itself.

In addition, each case study consists of a zip folder containing data in csv files and a document with further important information, such as the description of the data set. The zip folder is attached to the course material.

The coursebook and more in-depth information from the Internet or suitable reference books serve as basis. For example, data science community websites, such as **Kaggle** (<https://www.kaggle.com/>), provide a rich collection of freely available datasets and public notebooks. It is expected that you research further relevant literature for the chosen topic and use that as basis for the elaboration. Correctly reference them in the bibliography.

Especially the case studies in the module "Model Engineering" consist of conceptual tasks as well as programming tasks. Incorporate your findings and visualizations into a final document. For programming, the python and R languages are recommended because they are now the most common programming languages in the Data Science community. In addition, the freely available web interface **Jupyter Notebook** (<https://jupyter.org/>) is very helpful, especially for a first, exploratory investigation of the dataset. In this web interface, program parts, visualizations, and text can be integrated into a single notebook.

Do not forget to include your well-documented code as an attachment in the final submission, as it is possible to submit one document via Turnitin only.

Note on copyright and plagiarism:

Please take note that IU Internationale Hochschule GmbH holds the copyright to the examination tasks. We expressly object to the publication of tasks on third-party platforms. In the event of a violation, IU Internationale Hochschule is entitled to injunctive relief. We would like to point out that every submitted written assignment is checked using a plagiarism software. We therefore suggest not to share solutions under any circumstances, as this may give rise to the suspicion of plagiarism.

1.1 Task 1: Development of an Interpretable Model for Breast Cancer Prediction

Background: According to the breast cancer is one of the most common cancer types overall. Diagnosing a breast pathology in time is very important since it significantly increases the chances of successful treatment and survival. Thus, a fast and accurate classification of a tumor as benign or malignant is important. Moreover, to increase technology acceptance, the trustworthiness of the approach is critical.

A group of oncologists have seen impressive breast cancer identification results from an AI algorithm on a fair. However, the group did not understand why the model predicted certain tumors as malign and others as benign - the validity of the model was questioned in some cases. The group also discussed the most important features for the predictions. Finally, they decided to ask a group of data scientists, you are one of them, for possibilities to understand the prediction of a possible machine learning model. Your task is to develop a classification model that predicts a tumor to be malignant or benign with high accuracy (F1 score > 0.95). Moreover, the model should be interpretable. The outcomes should be presented/communicated to non-experts to convince them about the trustworthiness of the chosen approach.

Task: Support a group of oncologists with the interpretable prediction model to allow for additional indications that can be produced automatically as well as support understanding to ease technology acceptance.

Dataset: The dataset and additional information are provided in the file:
“Additional_Information_Case_Study_DLBDSE01”

Detailed description of the task: The task consists of both a programming part and conceptual subtasks. In the following, find a description of different points to be considered in the final document:

- Organize the project using the CRISP-DM or the MS Team Data Science method. Make a proposal on how to build the folder structure of a Git repository for the project.
- Assess the quality of the data set provided. Prepare and visualize your findings in such a way that business partners can understand the important relationships in a clear and simple way.
- Establish a collection of candidate models that meets the requirements and investigate feature importance on different models. (**Tip:** consider the interpretability of the models and increase complexity gradually. The accuracy should be high, but the interpretability of the model is the final aim.)
- To help business partners develop confidence in your new model, discuss the importance of each explanatory variable and make the model results as interpretable as possible. In addition, a detailed error analysis is very important so that the business partners also understand the weaknesses of your approach.

In the last step of the project, propose how your model can be integrated into the daily work of the doctors, for example how a graphical user interface (GUI) could look like.

Note: There is not one perfect solution. It is important to keep the concepts and techniques you have learned during the course in mind and to carefully prepare and process the data accordingly. At the end of each step of this use case, you should critically assess whether all necessary operations have been conducted and argue for the decisions you have made during this process.

1.2 Task 2: Development of a Demand Forecasting Model for Public Transport

Background: As one of the data scientists in an American public transport company, the logistic team ask you for help. They want to optimize the usage of the company's resources (the vehicles/taxis) by covering places with high demand for public transport. You should help them to predict the most important clusters for public transport (using taxis) as well as their progression over the day. To this end the logistic team gives you a dataset of the GPS data and starting time of the company's vehicles over the last six months. Use this data to create a regression model, possibly multiple ones, to predict the average demand of taxis per hour over the day in the ten most important clusters in the city. Discuss possible demand variability and propose a suitable prediction approach.

Task: Support the logistic team in the company with a prediction approach for the average demand of taxis per hour over the day in the ten most important clusters of the available data.

Dataset: The dataset and additional information are provided in the file:

"Additional_Information_Case_Study_DLBDSE01"

Detailed description of the task: The task consists of both a programming part and conceptual subtasks. In the following, find a detailed description of open questions to be answered in the final document:

- Organize the project using the CRISP-DM or the MS Team Data Science method. Make a proposal on how to build the folder structure of a Git repository for the project.
- Assess the quality of the data set provided. Prepare and visualize the data in such a way that business partners can understand the important relationships in a clear and simple way.
- **Tip:** Use a clustering approach, e.g. (K-Nearest-Neighbor) to compute cluster-centers over the complete dataset; train a regression model per cluster to predict the mean number of cars in the cluster area (area is defined via the closest cluster mean) per time of the day (per hour).
- To help business partners develop confidence in your new model, a detailed error analysis (e.g., Are there seasonal changes? Is one regression model sufficient?) is very important so the business partners understand the weaknesses of your approach.

In the last step of the project, propose how your model can be integrated into the daily work of the logistic team, for example how a graphical user interface (GUI) could look like and how new data can be integrated as part of a continuous integration process.

Note: There is not one perfect solution. It is important to keep the concepts and techniques you have learned during the course in mind and to carefully prepare and process the data accordingly. At the end of each step of this use case, critically assess whether all necessary operations have been conducted and argue towards the decisions you have made during this process.

1.3 Task 3: Development of a Spam Filter

Background: In recent years the short messages communication has become ubiquitous for social interactions around the globe. This applies both to private as well as professional sectors. The amount of spam messages has increased as well, particularly on publicly available channels. Your company would like to install an open channel for some of their products to enable fast and interactive feedback for their customers or possible future customers. You are the data scientist in the company asked to support the service team with building a spam filter for this channel.

Therefore, you are provided with a dataset compiled from different open-source datasets for short messages. One request from the service team is that the model should allow for adjustments with respect to the risk of allowing spam to pass, due to the huge variety of message quality that the (possible) customers provide. Therefore, the spam filter should be adjustable via different spam-risk levels, e.g., „low-risk“ (very restrictive) and „high-risk“ (not restrictive). The messages passing the „low-risk“-level are immediately displayed. The other ones are moved to a folder for further analysis. You should also provide a strategy for further analysis, so as not to lose customers.

Task: Support the service team via a classification model that can identify spam from short messages and that allows for an adjustable risk level and a plausible strategy to deal with erroneous spam classification.

Dataset: The dataset and additional information are provided in the file:

“Additional_Information_Case_Study_DLBDSE01”

Detailed description of the task: The task consists of both a programming part and conceptual subtasks. Here follows a detailed description of open questions to be answered in the final document:

- Organize the project using the CRISP-DM or the MS Team Data Science method. Make a proposal on how to build the folder structure of a Git repository for the project.
- Assess the quality of the data set provided. Prepare and visualize your findings in such a way that business partners can understand the important relationships in a clear and simple way.
- Create, train, test a binary classification model with the provided dataset.
(**Tip:** due to the short length of the messages, word counts, and simple models might already perform reasonable)
- To help business partners develop confidence in your new model, a detailed error analysis is very important so that the business partners also understand the weaknesses of your approach.

In the last step of the project, you should propose how your model can be integrated into the daily work of the service team, for example how a graphical user interface (GUI) could look like.

Note: There is not one perfect solution. It is important to keep the concepts and techniques you have learned during the course in mind and to carefully prepare and process the data accordingly. At the end of each step of this use case, you should critically assess whether all necessary operations have been conducted and argue for the decisions you have made during this process.

2. ADDITIONAL INFORMATION FOR THE EVALUATION OF THE CASE STUDY

When conceptualizing and writing the case study, the evaluation criteria and explanations given in the writing guidelines should be considered.

Capture of problem: How is the problem from a business area translated into a Data Science project, i.e., which Data Science use case can be derived from the problem?

Concept: Apply and incorporate the relevant concepts of the course into the study. Examples include:

- Performing exploratory data analysis, such as calculating correlations.
- What models are used for prediction (e.g., linear regression, decision tree procedures...)?
- What techniques are used to select the explanatory variables ("features")?
- How to avoid overfitting of the system (e.g., by training with several hyperparameters and applying regularization methods, ...)?

Analysis: Provide an accurate and detailed analysis of the data and the modeling approach.

Conclusion: Describe the main results of your study in a clear and quantitative way. Take the same structured approach as you did in analyzing the data and modeling approach. Answer questions that go beyond just modeling, such as, "How can the model be used in daily work?", "Are further developments necessary to move the model into an operational mode?"

3. TUTORIAL SUPPORT

Students have the option to make use of any one of several opportunities to get support for their case study analysis with the course tutor. Taking advantage of these opportunities is the responsibility of the student and the use of these services is voluntary. It is possible to contact the tutor regarding formal and general questions about working on the case study. Please note: a review of outlines and aspects of the presentation is not intended here, since the student's ability to work independently is part of the evaluation and counts as a part of the overall assessment. There are, however, general tips for developing the case study to help you get started.